INTEGRATED GENE–PROTEIN–PATHWAY ANALYSIS

Integrative Genomics and Proteomics of Alzheimer's Disease

How Alzheimer's genetic risk loci relate to protein abundance in the AD brain, where they converge on shared pathways, and which hypotheses could establish cause.

Summary

Of 89 prioritized candidate genes at 75 Alzheimer's risk loci, 13 encode proteins that are differentially abundant in AD brain. That rate (25% of measured genes) is no higher than for random genes (21%; Fisher p = 1.0), so overlap alone does not show that risk genes act through protein levels.

75
risk loci from 84 lead variants (Bellenguez 2022 + APOE)

89
prioritized candidate genes

13
candidates with altered brain protein (FDR < 0.05)

43%
candidates never measured in brain proteomics

- **Strongest gene–protein overlaps:** APP, CLU and APOE are raised in AD cortex on two or three proteomic platforms. BIN1, FERMT2 and TPCN1 are lowered in the largest dataset (TMT, ~400 brains).
- **Shared pathway:** regulation of amyloid-β formation and clearance (GO, FDR 3×10^{-6}) is the one process where genetic candidates and changed proteins converge (APOE, BIN1, CLU).
- **Biggest gap:** 38 of 89 candidates were never quantified in bulk brain proteomics, including most microglial risk genes (TREM2, SPI1, MS4A4A, MS4A6A, CR1, LILRB2, SIGLEC11).
- **Interpretation:** most overlapping changes are better explained as downstream of amyloid pathology or neuron loss than as variant effects. Six hypotheses below say how to tell these apart.

Data and methods

The analysis joins one GWAS and one brain proteomics resource at the gene level. Everything runs in Python from the files listed under Reproducibility.

Step	Source / method	Notes
GWAS loci	Bellenguez et al. 2022, *Nat Genet* (GWAS Catalog GCST90027158); stage I: 39,106 clinically diagnosed + 46,828 proxy cases vs 401,577 controls, European ancestry	83 lead variants from paper Tables 1–2 (GRCh38), plus APOE rs429358, which the tables exclude
Locus definition	Lead variants within 500 kb merged	84 variants → 75 loci
Gene mapping	Ensembl 111 protein-coding genes within ±250 kb (window set); nearest gene; paper-reported gene; published prioritization (paper tier 1/2 genes and consensus genes at known loci)	656 window genes (540 with pathway annotation); 89 prioritized genes
Proteomics	AMP-AD Agora AD vs control results: TMT (ROSMAP DLPFC, ~400 brains, adjusted for sex and post-mortem interval); LFQ (>500 brains, 4 regions); targeted SRM (DLPFC)	Differentially abundant (DA) = FDR < 0.05; TMT is the primary call

Step	Source / method	Notes
Background	40 random protein-coding genes outside all GWAS windows (seed 2026), looked up the same way	38 had Agora records
Statistics	Fisher's exact test for detection and DA rates; hypergeometric pathway test (GO BP 2021, KEGG, Hallmark; 15,492-gene universe); Benjamini–Hochberg FDR	Gene sets of 5–500 genes, ≥3 hits

Access limitation. The workspace could not download from GWAS Catalog, Open Targets, GEO, PRIDE or Synapse, so Agora's public gene API was read one gene at a time. That gives exact statistics for the 129 genes queried but not the full proteome, which limits the background to 38 genes. Open Targets locus-to-gene scores were not used.

GWAS loci and candidate genes

The 84 lead variants collapse into 75 loci holding 656 protein-coding genes within ±250 kb; 89 genes (including long-range candidates) carry published prioritization evidence.

- **Position check:** every paper-reported gene lies within 100 kb of its lead variant except three gene-desert loci (CLNK 339 kb, COX7C 306 kb, SEC61G 114 kb). Those labels are nearest-gene names, not functional calls.
- **Merged loci:** TREM2/TREML2/UNC5CL (chr6p21), CLU/PTK2B (chr8p21), and both SORL1, PLCG2, MME and SLC24A4 signals each form one locus.
- **Labels that hide the likely gene:** EED (consensus PICALM), SPDYE3 (ZCWPW1/NYAP1/PILRA), MINDY2 (ADAM10), WNT3 (MAPT region), BCKDK (KAT8). These carry literature, not computed, prioritization.
- **No protein-coding gene:** the two IGH-cluster loci (chr14q32) map to immunoglobulin segments only.

Gene–protein overlap

Thirteen GWAS candidates are differentially abundant in AD brain: 7 in the primary TMT dataset and 6 more only in LFQ or SRM. GWAS candidates are not over-represented among changed proteins.

Test	GWAS candidates	Random genes	Fisher OR (p)
Quantified in TMT	50 / 89 (56%)	19 / 38 (50%)	1.28 (0.56)
DA in TMT, of quantified	7 / 50 (14%)	3 / 19 (16%)	0.87 (1.0)
DA on any platform, of quantified	13 / 51 (25%)	4 / 19 (21%)	1.28 (1.0)

The background is small, so these tests can only rule out a large enrichment. Two random genes (GNG12, C18orf32) are as strongly changed as CLU, which shows how common AD-associated protein shifts are.

Candidate genes with altered brain protein (log2 fold change, AD vs control; FDR in brackets)

Gene (locus label)	TMT DLPFC	Best LFQ region	SRM DLPFC	Agreement
APP (APP)	+0.66 (3×10^{-16})	Aβ peptide, DLPFC +1.77 (8×10^{-12})	+5.76 (8×10^{-12})	Up, 3 platforms

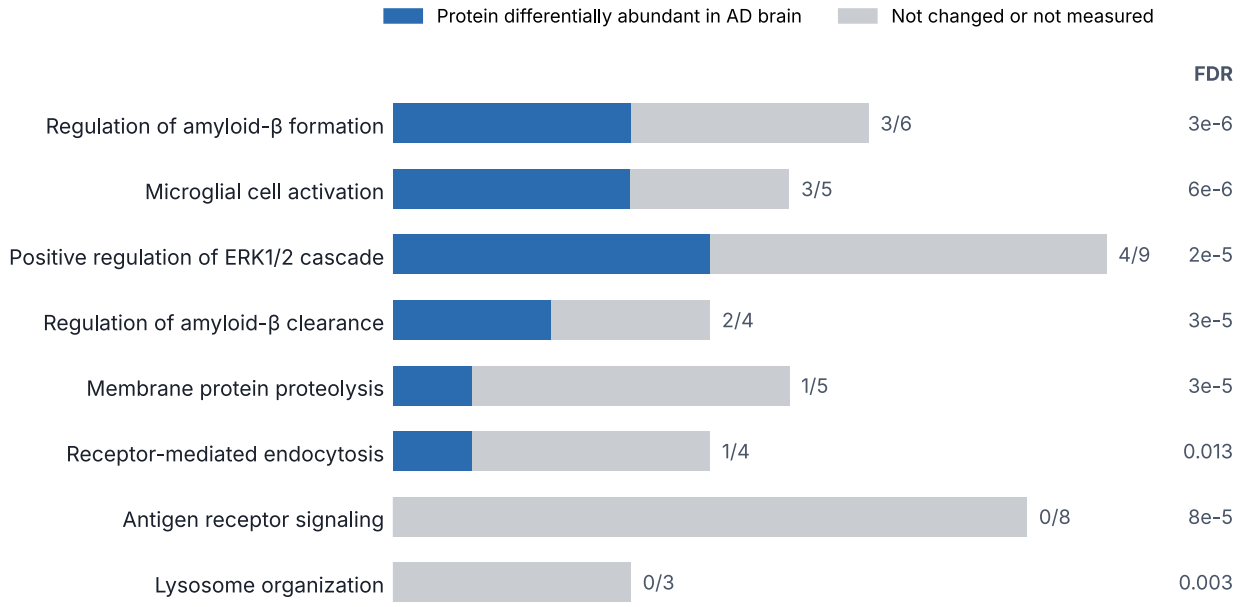
Gene (locus label)	TMT DLPFC	Best LFQ region	SRM DLPFC	Agreement
CLU (CLU)	+0.10 (0.002)	DLPFC +0.23 (5×10^{-7})	+0.25 (1×10^{-11})	Up, 3 platforms
APOE (APOE)	+0.11 (0.048)	DLPFC +0.18 (0.006)	+0.12 (0.009)	Up, 3 platforms
BIN1 (BIN1)	−0.06 (0.020)	DLPFC +0.07 (0.015)	−0.06 (0.0005)	Down in TMT and SRM; LFQ opposite
FERMT2 (FERMT2)	−0.06 (0.006)	n.s.	n.s.	Down, TMT only
TPCN1 (TPCN1)	−0.09 (0.008)	not quantified	not quantified	Down, TMT only
WDR81 (WDR81)	+0.03 (0.035)	not quantified	not quantified	Up, TMT only, small
ICA1 (ICA1)	−0.04 (0.050)	DLPFC −0.15 (0.0006)	not quantified	Down, LFQ
PTK2B (PTK2B)	n.s.	DLPFC −0.11 (0.0008)	n.s.	Down, LFQ
MAPT (WNT3)	n.s.	isoform, DLPFC +0.39 (1×10^{-6})	n.s.	Up, LFQ isoforms
ADAM10 (MINDY2)	n.s.	DLPFC +0.29 (0.0003)	not quantified	Up, LFQ
COX7C (COX7C)	n.s.	DLPFC −0.14 (0.031)	not quantified	Down, LFQ
CTSB (CTSB)	n.s.	AntPFC +0.13 (0.042)	not quantified	Up, LFQ, 1 of 4 regions

The remaining 38 quantified candidates (e.g. SORL1, PICALM, CD2AP, PLCG2, INPP5D, ABCA7, SORT1, GRN, TMEM106B) show no AD-associated change at FDR < 0.05.

Overlapping pathways

Amyloid-β production and clearance is the clearest point where genetic risk and changed brain proteins meet. Immune signalling is genetically enriched but has no matching protein change in bulk cortex.

Changed proteins concentrate in amyloid-handling processes



GO Biological Process 2021 enrichment of 89 prioritized GWAS genes (hypergeometric, BH FDR). Bar length = GWAS genes in the process; blue = protein also differentially abundant in AD brain (AMP-AD Agora, FDR < 0.05).

- **Converging:** APOE, BIN1 and CLU sit in both “regulation of Aβ formation” and the DA set; APP, CLU and MAPT link to microglial activation; APOE, APP, FERMT2 and PTK2B to ERK1/2 signalling.
- **Genetics only:** antigen-receptor signalling (BLNK, HLA-DQA1/DRB1, INPP5D, PLCG2, LIME1, PLEKHA1, RBCK1) and lysosome organization (ABCA1, GRN, TMEM106B) have no DA proteins.
- **Unbiased check:** using all 540 annotated window genes, the top processes are immune: complement regulation (6 genes, 2 loci), antigen-receptor signalling (21 genes, 12 loci, FDR 4×10^{-4}) and neutrophil-mediated immunity (36 genes, 21 loci, FDR 0.003). Amyloid processes do not lead there, because the prioritized list is partly curated from amyloid biology.

> **Caution.** 255 of 297 tested GO/KEGG sets pass FDR < 0.05 for the prioritized genes. That inflation reflects how the list was built, so treat these enrichments as descriptive, not as independent evidence.

Testable hypotheses

Each hypothesis separates “the variant changes the protein” from “the disease changes the protein”; overlap alone cannot do that.

H1 CLU is raised by pathology, not by its risk variant.

Prediction: the lead allele (rs11787077) has no brain or CSF pQTL for clusterin, and GWAS and pQTL signals do not colocalize (PP.H4 < 0.5). The CLU increase shrinks after adjusting for plaque burden.

Test: colocalization and Mendelian randomization with ROSMAP brain pQTLs and plasma/CSF pQTLs; regression of CLU abundance on genotype and amyloid score.

Falsified if: coloc PP.H4 > 0.8 with the protective allele shifting CLU consistently across tissues.

H2 APOE protein rise tracks ε4 dose and plaque deposition.

Prediction: within AD and within controls, APOE abundance differs by ε4 copy number; the AD-vs-control difference falls when amyloid load is a covariate.

Test: stratify ROSMAP TMT/SRM APOE by genotype; compare insoluble vs soluble fractions.

H3 The BIN1 decrease is a loss of the neuronal isoform.

Prediction: peptides unique to neuronal BIN1 (neuron-specific exons such as the CLAP domain) fall in AD while ubiquitous-isoform peptides do not, explaining the LFQ–TMT sign conflict; the rs6733839 risk allele shifts the isoform ratio.

Test: peptide-level re-analysis of the raw TMT/LFQ data; single-nucleus RNA isoform analysis by genotype.

H4 TPCN1 is the effector gene at the chr12q24 locus and acts through endolysosomal Ca^{2+}.

Prediction: rs6489896 colocalizes with a TPCN1 eQTL/pQTL in brain or microglia; TPCN1 knockdown in iPSC neurons or microglia changes Aβ secretion or endolysosomal pH.

Test: coloc against brain and microglia QTL panels; CRISPRi of the lead enhancer, then TPCN1 protein and Aβ40/42.

H5 Microglial risk genes act on abundance that bulk cortex cannot see.

Prediction: TREM2, MS4A4A/6A, SPI1, CR1 and LILRB2 show genotype-dependent levels in sorted microglia or CSF (e.g. soluble TREM2 by MS4A genotype), even though bulk tissue does not detect them.

Test: targeted PRM assays in microglia-enriched fractions; CSF proteomics with genotype data.

H6 Synaptic and mitochondrial decreases (PTK2B, ICA1, COX7C) reflect neuron loss.

Prediction: the decreases disappear after adjusting for neuronal marker abundance or estimated cell-type proportions.

Test: cell-type deconvolution of the proteomic data, then re-run the AD contrast.

Falsified if: decreases persist and associate with lead-variant genotype.

Evidence gaps and limitations

The largest gap is measurement: 43% of candidate genes have no brain protein data, and none of the overlaps carry genotype-level evidence.

Gap	What it affects	What would close it
38 of 89 candidates not quantified (mostly microglia-specific or low-abundance: TREM2, SPI1, MS4A4A/6A, CR1, LILRB2, SIGLEC11, RHOH, BLNK, ABCA1, MME, EPHA1)	Immune loci cannot be tested at protein level	Targeted PRM/SRM; sorted-cell or CSF/plasma proteomics
No variant-to-protein link (pQTL, colocalization, MR)	Overlap cannot be read as causal	Brain/CSF pQTL coloc and MR (H1, H4)

Gap	What it affects	What would close it
Bulk tissue, cross-sectional, post-mortem	Changes mix cell loss, gliosis and pathology	Cell-type deconvolution; adjustment for plaque/tangle scores
Background of 38 random genes	Enrichment test only detects large effects	Full proteome table (Agora/Synapse download)
Single cortical region for TMT (DLPFC)	Regional effects (e.g. hippocampus) missed	Multi-region datasets; LFQ regions only partly fill this
Prioritized gene list partly literature-curated	Pathway enrichment is circular	Data-driven mapping: Open Targets L2G, eQTL/sQTL coloc, fine-mapping
Gene-level summaries read through a web reader	Small risk of transcription error	Re-run scripts on downloaded tables; raw lookups saved for audit

All DA calls are AD vs control associations. A protein can change in AD because of the disease, treatment, agonal state or tissue composition, so no overlap here shows that a risk variant changes a protein.

APPENDIX

Methods walkthrough

Twelve steps take the analysis from GWAS loci to testable hypotheses. Code: `scripts/01_map_loci.py` (steps 2–4) and `scripts/02_integrate.py` (steps 5–9).

1 Frame the question

Are Alzheimer's risk genes also proteins that change in AD brain, and where do genetic and protein evidence meet in shared pathways? The design is gene-level integration: GWAS loci → candidate genes → protein changes → pathways. Overlap is a lead to test, never proof that a variant causes a protein change.

2 Collect GWAS results

- **Study:** Bellenguez et al. 2022, *Nature Genetics*; GWAS Catalog GCST90027158.
- **Sample:** stage I, 39,106 clinically diagnosed and 46,828 proxy cases vs 401,577 controls, European ancestry.
- **Extracted:** 83 genome-wide significant lead variants from paper Tables 1–2 (rsID, chromosome, GRCh38 position, OR, P, reported gene), plus APOE rs429358, which the tables exclude.
- **Check:** values spot-checked against the GWAS Catalog record (e.g. BIN1 rs6733839, P = 6×10^{-118}).
- **With full summary statistics:** clump by LD instead (PLINK `--clump`, $r^2 < 0.1$, 1 Mb).

3 Define loci

Lead variants less than 500 kb apart on one chromosome were merged: 84 variants → 75 loci. TREM2/TREML2/UNC5CL, CLU/PTK2B and the two SORL1 signals each form one locus.

4 Map loci to candidate genes

Gene coordinates come from Ensembl 111 (protein-coding genes only).

Layer	Rule	Count
Window	Any gene within ±250 kb of a lead variant	656 genes
Nearest	Gene body closest to the lead variant	1 per locus
Reported	The paper's locus label	75 labels
Prioritized	Paper tier 1/2 genes plus long-standing consensus genes at known loci	89 genes

- **Position check:** every reported gene lies within 100 kb of its lead variant except three gene-desert loci (CLNK 339 kb, COX7C 306 kb, SEC61G 114 kb).
- **Labels that hide the likely gene:** EED → PICALM, MINDY2 → ADAM10, WNT3 → MAPT, SPDYE3 → ZCWPW1/PILRA; the consensus gene was added to the prioritized set.

5 Get brain proteomics data

- **Source:** AMP-AD Agora, AD vs control results.
- **TMT (primary):** ROSMAP dorsolateral prefrontal cortex, ~400 brains, adjusted for sex and post-mortem interval.
- **LFQ:** >500 brains, 4 cortical regions. **SRM:** targeted assays, DLPFC.
- **DA call:** differentially abundant if FDR < 0.05.
- **Comparison set:** the full proteome could not be downloaded, so each gene was queried separately: 89 prioritized genes plus 40 random protein-coding genes outside all GWAS windows (seed 2026). Raw results: `data/agora_raw_lookup.txt`.

6 Test the overlap

Fisher's exact test compares GWAS genes with random genes:

```
fisher_exact([[gwas_DA, gwas_notDA], [bg_DA, bg_notDA]])
```

Results are in the overlap table in the main report: GWAS genes are not enriched among changed proteins (DA in TMT 14% vs 16%; any platform 25% vs 21%). Protein changes are common in AD brain; two random genes (GNG12, C18orf32) changed as much as CLU.

7 Assign evidence tiers

Tier	Meaning	Genes
A	DA in TMT	APP, CLU, APOE, BIN1, FERMT2, TPCN1, WDR81
B	DA only in LFQ or SRM	ICA1, PTK2B, MAPT, ADAM10, COX7C, CTSB
C	Measured, not changed	38 genes, e.g. SORL1, PICALM, PLCG2, INPP5D
D	Not measured in brain	38 genes, e.g. TREM2, SPI1, MS4A4A, CR1

APP, CLU and APOE rise in AD on all three platforms. BIN1 and FERMT2 fall in TMT; BIN1 also falls in SRM but rises in LFQ.

8 Pathway enrichment

```
hypergeom.sf(k - 1, N_universe, n_pathway, n_query)
```

- **Test:** hypergeometric against GO Biological Process 2021, KEGG and Hallmark; universe 15,492 annotated protein-coding genes; Benjamini–Hochberg FDR; sets of 5–500 genes with ≥3 hits.
- **Prioritized genes (89):** top hit is regulation of amyloid-β formation (FDR 3×10^{-6}). This is partly circular: 255 of 297 tested sets pass FDR < 0.05 because the list was curated from the literature.
- **Window genes (540 annotated, unbiased check):** top hits are immune — antigen-receptor signalling (21 genes, 12 loci) and neutrophil-mediated immunity (36 genes, 21 loci).

9 Find overlapping pathways

- **Converging (risk genes and changed proteins):** Aβ formation (APOE, BIN1, CLU); microglial activation (APP, CLU, MAPT); ERK1/2 signalling (APOE, APP, FERMT2, PTK2B).
- **Genetics only:** antigen-receptor signalling (HLA, INPP5D, PLCG2, BLNK); lysosome organization (GRN, TMEM106B, ABCA1).

10 Turn overlaps into hypotheses

Each hypothesis separates "the variant changes the protein" from "the disease changes the protein" (full predictions and tests in the main report, H1–H6).

1. CLU is raised by pathology, not its risk variant — pQTL colocalization, Mendelian randomization, plaque-load adjustment.
2. APOE protein tracks ε4 dose and plaques — stratify by genotype.
3. BIN1's decrease is loss of the neuronal isoform — peptide-level isoform analysis.
4. TPCN1 is the causal gene at its locus and acts through endolysosomal Ca^{2+} — QTL colocalization, CRISPRi in iPSC cells.
5. Microglial risk genes change in microglia, invisible in bulk tissue — sorted-cell or CSF proteomics.
6. Synaptic and mitochondrial decreases (PTK2B, ICA1, COX7C) reflect neuron loss — cell-type adjustment.

11 Record evidence gaps

- 43% of candidates (38 of 89) were never measured in brain proteomics.
- No variant-to-protein link yet (pQTL, colocalization, MR).
- Bulk, post-mortem data; TMT from one region only.
- Small comparison set (38 random genes).

12 Upgrade with full data

1. Download "TMT Proteomics for Agora" from Synapse and use every measured protein as background.
2. Replace the curated gene list with Open Targets locus-to-gene scores or eQTL/pQTL colocalization (`coloc` in R).
3. Run Mendelian randomization from brain or CSF pQTLs to AD risk (`TwoSampleMR`).

Reproducibility

All scripts, inputs, raw proteomics lookups and result tables are in `ad_integrative_analysis.zip`. Two commands rerun everything:

```
python scripts/01_map_loci.py data/ensembl111_hg38_genes.rda data/bellenguez2022_lead_variants.csv
results
python scripts/02_integrate.py . data/ensembl111_hg38_genes.rda
```

- `01_map_loci.py` — merges lead variants into loci and maps genes (window, nearest, reported, prioritized).

- `02_integrate.py` — parses proteomics, runs overlap tests and pathway enrichment, and writes `integrated_gene_table.csv` (evidence tiers A–D for all 89 genes).

Sources

- Bellenguez C et al. New insights into the genetic etiology of Alzheimer's disease and related dementias. *Nat Genet* 2022. Open-access copy; GWAS Catalog GCST90027158
- AMP-AD Agora gene API, e.g. CLU record; Agora data and methods
- Gene sets: GO BP 2021, KEGG and Hallmark files from the GSEApy repository
- Gene coordinates: Ensembl 111 via the CRAN package enshuman; IDs via babelgene